Question 1:
Is bridge an intelligent device? If yes then how.
Answer:
Yes, a bridge operates at Layer 2, learns MAC addresses, and forwards frames only to the intended segment, reducing traffic.

Question 2:
What are the services provided by TCP?
Answer:
Reliability, flow control, error detection, connection-oriented communication, congestion control.

Question 3:
What functions does IP multicast abstraction allow an application to do?
Answer:
Send data simultaneously to multiple receivers efficiently without separate copies.

Question 4:
What is Zero-compression in IPv6?
Answer:
Replacing consecutive zeros in an IPv6 address by "::" to shorten it.

Question 5:
Which routing algorithm provides periodic routing table updates? Why?
Answer:
Distance Vector routing; it shares routing updates periodically with neighbors.

Question 6:
How does Host F identify the entire original datagram sent by Host B?
Answer:
Using the IP header fields: identification, fragment offset, and more fragments flag.

Question 7:
Is TCP an end-to-end protocol? Explain.
Answer:
Yes, TCP ensures reliable communication between source and destination hosts directly.

Question 8:
Who propagates multicast routing? Example?
Answer:
Multicast routers; example: Protocol Independent Multicast (PIM).

Question 9:
What data is inside a frame? How does the receiver know?
Answer:
Data packets or segments; identified by the Frame Type field.

Question 10:
How does IP reassemble out-of-order fragments?
Answer:
Using fragment offsets to place each fragment correctly, reassembling when all arrive.

Question 11:
Describe UDP endpoint identification with port numbers.
Answer:
Uses source and destination port numbers to identify processes on hosts.

Question 12:
Why is a server needed? Example?
Answer:
To provide centralized resources/services, e.g., web server delivering web pages.

Question 13:
Methods to span a bridge network over long distances?
Answer:
Leased lines and wireless links.

Question 14:
ARP technique where protocol addresses are independent from hardware addresses?
Answer:
Proxy ARP.

Question 15:
Which NAT type fails when an app uses IP address instead of domain name?
Answer:
Application Layer Gateway (ALG) dependent NAT.

Question 16:
Most used gateway protocol as EGP and two properties?
Answer:
BGP; path vector protocol, policy-based routing.

Question 17:
Purpose of extra characters between frames? Two problems?
Answer:
Define frame boundaries; problems: bit stuffing complexity and synchronization issues.

Question 18:
Three error types Internet layer detects?
Answer:
Checksum errors, unreachable destination, TTL expired.

Question 19:
NAT alternatives?
Answer:
Static NAT, Dynamic NAT, Port Address Translation (PAT).

Question 20:
Characteristics making IP multicast easy and unique?
Answer:
One-to-many delivery, group management, scalability, bandwidth efficiency.

Question 21:
Five disadvantages of point-to-point communication?
Answer:
High cost, complex, poor scalability, single points of failure, hard management.

Question 22:
How does a system deal with incoming ARP messages?
Answer:
Check if request is for own IP, reply with MAC, update cache on reply, ignore others.

Question 23:
TCP's process to establish reliable connection?
Answer:
Three-way handshake: SYN, SYN-ACK, ACK.

Question 24:
Why does EGP not use routing metrics? Example?
Answer:
Uses policy-based routing for administrative control, e.g., BGP.

Question 25:
Classes of network based on mobility?
Answer:
Fixed, wireless, mobile networks.

Question 26:
ARP technique with hardware address smaller than protocol address?
Answer:
Inverse ARP.

Question 27:
Protocols for internal and external routing in autonomous groups?
Answer:
Interior Gateway Protocols (IGPs), Exterior Gateway Protocols (EGPs).

Question 28:
Protocol capable of buffering incoming and outgoing data both ways?
Answer:
TCP.

Question 29:
Do all LAN technologies include a type field in frames? Why/why not?
Answer:
No; some use length field, some have different frame formats.

Question 30:
How does Host F identify lost fragments of IP datagram?
Answer:
Missing fragment offset values and timeout detection.

Question 31:
Three main drawbacks of RIP?
Answer:
Limited hop count (15), slow convergence, routing loops.

Question 32:
TCP procedure for packet loss and consequences if neglected?
Answer:
Retransmission on timeout or duplicate ACKs; otherwise data loss and poor performance.

Question 33:
Technique to resolve reserved words in character stuffing?
Answer:
Bit stuffing.

Question 34:
How fragmentation works with different MTU sizes?
Answer:
Split packets into fragments with offset and flags for reassembly.

Question 35:
Five multicast routing protocols?
Answer:
PIM, DVMRP, MOSPF, CBT, SSM.

Question 36:
Can global Internet routers use static routing? Routing table size?
Answer:
No; dynamic routing used; tables have hundreds of thousands of entries.

Question 37:
Four factors to classify computer networks?
Answer:
Geographical scope, topology, transmission media, protocol.

Question 38:
Is ARP message format used for all frame formats?
Answer:
No; limited to specific frame formats like Ethernet.

Question 39:
NAT type failing with IP address usage instead of domain name?
Answer:
ALG based NAT.

Question 40:
Protocols exchanging routing info in autonomous systems?
Answer:
IGPs (OSPF, RIP), EGPs (BGP).

Question 41:
TCP/IP address resolution algorithms?
Answer:
ARP, RARP, Inverse ARP.

Question 42:
Physical addressing in WAN; example (1,5)?
Answer:
Switch 1, port 5.

Question 43:
UDP's best effort service purpose and example?
Answer:
No delivery guarantee; useful for streaming like video conferencing.

Question 44:
Characteristics making IP multicast easy and unique?
Answer:
Efficient one-to-many delivery, group management, scalable distribution.

Question 45:
Network tech to overcome failure automatically?
Answer:
Mesh topology with redundant paths and Spanning Tree Protocol.

Question 46:
IPv6 boundary limits and special address types?
Answer:
No strict boundary like IPv4; Unicast, Multicast, Anycast.

Question 47:
How to process received ARP messages?
Answer:
Check request IP, reply if matches, update cache on reply, ignore others.

Question 48:
TCP reliable connection establishment steps?
Answer:
Three-way handshake: SYN (client), SYN-ACK (server), ACK (client).

Question 49:
Why EGP doesn't use routing metrics?
Answer:
Policy-based routing preferred over metrics; example: BGP.

Question 50:
Wiring scheme for a low-cost limited area network with reasons?
Answer:
Twisted pair; low cost, easy installation, flexible, adequate LAN speed.

Question 51:
Routing table for R2 in a 4-network 3-router Internet (simplified):
Hop count from 192.4.10.0 to 30.0.0.0 = 2 via R2 to R1.

Destination	Mask	Next Hop
30.0.0.0	255.0.0.0 (/8)	R1
40.0.0.0	255.0.0.0 (/8)	Direct
128.1.0.0	255.255.0.0 (/16)	R3
192.4.10.0	255.255.255.0 (/24)	Direct

Question 52:
What are the reasons for IPv6 adoption?
Answer:
1: Exhausted IPv4 addresses
2: Simpler Headers
3: Built-in security (IPsec)
4: Better Multicast
5: Stateless auto-configuration

Question 53:
Static routing in Global Internet?
Answer:
No, (too dynamic; e.g., BGP tables have ~900K routes).
Example: ISPs use BGP due to frequent toplogy changes.

Question 54:
Technique to handle reserved words in character stuffing?
Answer:
Byte Stuffing(e.g., ESC sequences)
Example: If FLAG is reserved, sender insert ESC before it.

Question 55:
What are the characteristics of making IP Multicast Unique?
Answer:
There are three characteristics of making IP Multicast Unique:

- Efficiency: Single stream to multiple receivers
- Scalability: Reduces bandwidth usage vs. Unicast
- Group Management: Protocols like IGMP manage members.

Question 56:
What are the alternatives to NAT?
Answer:
1: IPv6 (eliminates need for NAT due to large address space)
2. Proxy Servers(mediate connections without NAT)
3. Tunneling (e.g., VPNs)

Question 57:
What are the features and types of Resolution?
Answer:

Feature	Type Of Resolution
Useful with any hardware	T
Address change affects all hosts	T
Protocol address independent of hardware address	T, D
Hardware address must be smaller than protocol address	C
Protocol address determined by hardware address	C
Requires hardware broadcast	D
Adds traffic to a network	D
Produces resolution with minimum delay	T, C
Implementation is more difficult	D

Question No. 58:
Construct the table of Classes of Address?
Answer:

First Four Bits Of Address	Table Index (in decimal)	Class of Address
0000	0	A
0001	1	A
0010	2	A
0011	3	A
0100	4	A
0101	5	A
0110	6	A
0111	7	A
1000	8	B
1001	9	B
1010	10	B
1011	11	B
1100	12	C
1101	13	C
1110	14	D
1111	15	E

Question No. 59:
Define Broadcast, Multicast and Unicast?
Answer:
Broadcast: When an application Broadcasting data, it makes a copy of the data available to all other computer on the network.
Multicast: Special types of addresses in IPv6 used for multiple destinations possibly not at the same site. Are known as Multicast.
Unicast: 1-to-1 communication/connection that uses TCP(Transmission Control Protocol) for delivering systems over the Internet.

Question No. 60:
What are features of BGP?
Answer:
Type: EGP(used as between Autonomous systems)
Main Feature:
Enables routing between ASes.
Supports policy based routing
Uses TCP for reliable communication.

Provide AS path information for each destination.
Current standard Version: BGP-4
Support Transit Routing

Question No.61:
Change the following into Binary Form:

- **154.31.161.13**

10011010 00011111 10100001 00001101

- **202.32.15.7**

11001010 00100000 00001111 00000111

- **192.168.1.5**

11000000 10101000 00000001 00000101

Question No.62:
Network Engineer has three address resolution methods. How many methods does TCP/IP supports in real environment. Write the names of methods and support your answer with solid reasons?
Answer:
Address Resolution algorithm can be grouped into three basic categories:

- Table Lookup
- Closed-Form Computation
- Message Exchange

TCP/IP can use any 3 of them depending on the addressing scheme used by the underlying hardware.

Question No. 63:
Diff. between IGPs and EGPs? Give example of each?
Answer:
IGPs(Interior Gateway Protocols):
Uses among routers within an autonomous system. The destination lies within an IGP.
Example of IGPs: Routing Information Protocol(RIP), Open Shortest Path First Protocol(OSPF)
EGPs(Exterior Gateway Protocols):
Uses between autonomous system. The destination lies throughout the Internet.
Example of EGPs: Border Gateway Protocol(BGP)

Question No. 64:
Consider the IP Address 178.200.127.5 & its corresponding subnet mask 255.255.255.0 find the following:
A) No of bits used for subnetting
B) Total no of usable hosts in subnet
C) Write CIDR notation for the above scenario.
Answer:
Given IP: 178.200.127.5, Subnet Mask: 255.255.255.0
A) Bits usd for Subnetting:
The subnet mask 255.255.255.0 uses 24 bits for the network portion(no additional subnetting bits beyond default class B).
B) Total hosts in Subnet:
Host bits = 32-24 = 8 bits
Usable Hosts = 2^8 - 2 = 254
C) CIDR Notation: 178.200.127.5/24

Question No. 65:
In an organization where a large network list deployed & some remote function locations that can be used for network troubleshooting. Write name of the protocol any tools which uses this protocol. Also identify any two tools which uses this Protocol.
Answer:
Protocol: ICMP(Internet Control Message Protocol)
Tools: 1. Ping(uses ICMP Echo Request/Reply), 2. Traceroute(uses ICMP Time Exceeded Message)

Question No. 66:
Find the last assignable host address in block if one of subnet ID is 192-1-1-0/26.
Answer:
1) /26->Subnet mask

255.255.255.192-> host bit 32-36= 6 bits
2^6=64 address per subnet
2) Subnet 192-1-1-0/26 covers address 192.1.1.0 to 192.1.1.63
3) Network address = 192.1.1.0
4) Broadcast Address = 192.1.1.63
5) Last assignable host one less than broadcast = 192.1.1.62

Question No.67:
13. Diff types of ICMP messages are given below. Identify Error messages & Informational messages from the following.
Echo request/reply
Time exceeded
Destination unreachable
Addren Mask request/reply
Answer:
Error Messages:
- Time exceeded
- Destination unreachable

Informational Messages:
- Echo Request/Reply
- Address Mask Request/Reply

Question No. 68:
Write the name of routing metric used by Routing Information Protocol (RIP)?
Answer:
Hop Count Metrics: Numbers of routers to reach the destination.

Question No. 69:
Which types of NAT (Network Address Translation) devices are available?
Answer:
1) Basic NAT(1-to-1 mapping without post translation).
2) NAPT(Network address port translation)-> (aka PAT, overloads a single public IP with multiple private IP using ports)
3) Twice NAT (modifies both the source and destination addresses packets)

Question No. 70:
During transmission of data foron source to destination in a network of 50 computers, some packets contains errors. In this situation which field of IP datagram header will indicate about errors?
Answer:
Header Checksum field in IPv4 detect errors in the header/verifies integrity of the IP header.

Question No. 71:
Suppose that in computer network of chemical company, each packet is transferred with equal amount of time b/w systems considering this scenario what will be the jitter of the network.
Answer;
Jitter measures the path variation in the delay
If all packets have equal delay the the variation is O ms (Jitter -> 0ms) ->Final Answer of this question
If packets delays are inconsistent then the jitter is greater than 0 (Jitter = 1 ms / 2 ms).

Question No. 72:
Identify the type of delay when network components take time to forward data over the Network mediums
Answer:
Switching delay (It is the time required for network component (hub, bridge, packet switch) to forward data)

Question No. 73:
Write the Subnet Mask for IP address 110.0.1.2/23?
Answer:
/13: first 13 bits set to 1 in the mask

Binary: 11111111 11111000 00000000 00000000
Decimal: 255.248.0.0

Question No. 74:
Identify the type of Quality of Service (QoS) Specification of ATM.
Answer:
There are three specifications of ATM:

- Constant Bit Rate(CBR)[Used for audio and video].
- Variable Bit Rate(VBR) [Used for compressed Audio and Video].
- Available Bit Rate(ABR) [Used for typical data application(where the data rate may be unknown and bursty)].

Answer to this question is: **QoS Specification is done by Constant Bit Rate(CBR).**

Question No. 75:
Given IP Address 210.1.12/30 answer the following:
Number of bits used for Subnetting
Class of IP address
Write Subnet Mask
Answer:
Default Class for 210.x.x.x -> class{default mask/24}
Given: /30
So the surrounding bits = 30-24=6bits
Subnet Mask for /30 255.255.255.252

Bits for Subnetting	6
Class	C
Subnet Mask	255.255.255.252